
STRUCTURE, INFORMATION AND INFERENCE IN 5-A-SIDE FOOTBALL: CAN FORMATIONS COMPENSATE FOR LIMITED VISION?

Arpan Gupta
IIT Gandhinagar
Gandhinagar
24110051@iitgn.ac.in

Deep Buha
IIT Gandhinagar
Gandhinagar
24110082@iitgn.ac.in

Ramji Purwar
IIT Gandhinagar
Gandhinagar
ramji.purwar@iitgn.ac.in

ABSTRACT

We study cooperative multi-agent learning in a simplified 5-a-side football game with no goalkeepers, where every player sees the opponents and the ball only within a limited radius. Our team is controlled by a single parameter-shared reinforcement learning policy and plays against a hand-coded opponent that holds a formation and defends its own zone, so the opponent’s formation can be read from where its players stand. We ask three questions built around one theme, structure versus information: (Q1) can formation structure compensate for limited vision; (Q2) how quickly can the team infer that the opponent has changed formation, and how much does vision matter for that inference; and (Q3) can our team work out who an opponent playmaker with full-pitch vision is, and adapt to neutralize him, compared with a team unaware of him and an oracle told who he is. We formalize the problem as a decentralized POMDP, compare a recurrent policy with an explicit Bayesian belief filter and an oracle, define measures of adaptation, and describe the planned experiments.

Keywords Multi-agent reinforcement learning · Football · Partial observability · Formations · Opponent modelling

1 Introduction

1.1 Motivation

Team sports are rich everyday examples of multi-agent decision making. Each player acts on incomplete information (nobody sees the whole pitch at all times), coordinates with teammates without constant communication, and faces opponents who change tactics during the match. Teams cope in two ways. They rely on *structure*, such as a formation that tells every player where to be by default, and they gather *information*, by looking around and inferring what the opponent is doing. A coach who notices that the opposing team has switched from a defensive to an attacking shape, or that one opposing player reads the game unusually well, changes how the team plays. Building AI teams that can do the same is an open practical challenge: the same ingredients (partial information, cooperation, changing adversaries) appear in robot soccer [1], warehouse robotics, and multi-drone coordination.

1.2 The Game in Plain Words

Precise definitions follow in Sections 2 and 3; all numeric values are fixed later.

Setting. Two teams of five play on a rectangular pitch with no goalkeepers, for a fixed number of steps; more goals wins. One team is *ours*, and a single learned network is shared by all five players, each feeding it what it sees and receiving its own action. The *opponent* follows fixed hand-written rules and does not learn. A player can move, hold position, pass, dribble, shoot, tackle, or *shadow* a visible opponent (stay between him and our goal). Passes can be intercepted, tackles can fail, and shots score more often near goal and unmarked.

Formations and vision. A formation such as 2-2-1 (defenders–midfielders–forwards) gives each slot a home position, and both teams return to it after each goal. Depending on the experiment, the formation is *structural* (players play

worse far from home) or only a starting position. Each player sees opponents and the ball only within a radius, but always knows its teammates. The opponent keeps its shape: each player stays near home, presses the ball when it enters his zone, and otherwise drifts slightly toward the ball side. Its shape is therefore readable from where it stands, and it too sees only within a radius.

What changes during a match. (i) *Formation change (Q2)*: the opponent may switch shape mid-match, for example to attack when losing. Its players walk to new home positions, and our team is not told. (ii) *Playmaker (Q3)*: one opponent player may have full-pitch vision, seeing every player and passing lane and able to intercept anywhere. Our team is not told whether or who he is.

Learning. The team starts random, plays many matches, is rewarded for goals and winning (and lightly for passes and tackles), and is then tested in the situations above.

1.3 Problem Statement

We study a 5-a-side game without goalkeepers on a 2D pitch. Our team is one parameter-shared reinforcement learning policy. The opponent is a fixed, hardcoded, formation-holding zonal team. Every player sees opponents and the ball only within a limited radius. The project has one theme, *structure versus information*, and three questions:

1. **Q1 (structure compensates for information)**: Can formation structure compensate for limited vision? Does its benefit grow as the radius shrinks, and does the best formation depend on the radius?
2. **Q2 (inference)**: If the opponent changes formation, how quickly can our team infer it from partial observations, and how much does vision matter? We compare a recurrent policy, an explicit Bayesian belief filter, and an oracle told the true formation.
3. **Q3 (adapting to a playmaker)**: If one opponent has full-pitch vision, can our team identify him and neutralize him? How much lost win rate does an adaptive team recover, compared with an unaware team and an oracle told who he is?

In terms of the course, the project combines learning under limited observation (Direction D) with learning about unknown opponent behaviour against a full-information oracle (Direction C). The heuristic formation win matrix of Phase 0 is also a normal-form game between formations, whose best-response and maximin formations we compare with what the learned teams choose.

Known: the rules, our formation, teammates' positions, the set of possible opponent formations, and (oracle variants only) the true opponent formation and the playmaker's identity. **Unknown to our team:** the opponent's current formation, when it switches, whether and which opponent has full vision, and the payoffs of the formation game, which must be learned.

Contributions. A minimal 5v5 environment with formations, limited vision, a switching zonal opponent and an optional full-vision playmaker; a controlled measurement of the structure benefit across vision radii; a comparison of recurrent, belief-filter and oracle inference with delay, lag and regret measures; and a playmaker-identification study with a control that separates awareness from the extra shadow actions.

2 Preliminaries

2.1 Notation

Numeric values (pitch size, speeds, radii, horizon) are fixed during implementation.

2.2 Formations, Structure Levels and Positional Control

A **formation** $f \in \mathcal{F}$ assigns each player a **slot** k (defender, midfielder, forward) and an **anchor** $\alpha_f(k)$; player i occupies slot $k = i$. Formations are not enforced. The **structure level** ζ sets how much structure our team is given:

- $\zeta = 0$ (*none*): the formation only sets kickoff positions; anchors are not observed and there is no hold-shape action.
- $\zeta = 1$ (*structure as information*): the player observes its anchor and can hold shape, but leaving the anchor is free.
- $\zeta = 2$ (*structure as constraint*): as $\zeta = 1$, and effectiveness depends on distance from the anchor through the **control rating**

$$c_\zeta(x, \alpha) = \begin{cases} 1, & \zeta < 2, \\ c_{\min} + (1 - c_{\min}) \exp(-\|x - \alpha\|^2 / L^2), & \zeta = 2, \end{cases} \quad (1)$$

Table 1: Notation.

Symbol	Meaning
$\mathcal{N}, \mathcal{M}$ (each $\{1, \dots, 5\}$)	Players of our team, players of the opponent
$T, t \in \{0, \dots, T\}$	Horizon and time step
$\mathcal{P} \subset \mathbb{R}^2$	Pitch (bounded rectangle)
$x_i^t, y_j^t, b^t \in \mathcal{P}$	Positions of our player i , opponent j , and the ball
$\beta^t \in \{\text{ours}, \text{opp}, \text{loose}\}$	Possession state
$\mathcal{F}, \alpha_f(k)$	Set of formations; anchor (home) of slot k in formation f
$f, g^t \in \mathcal{F}$	Our formation (fixed); opponent formation at t (hidden)
$\zeta \in \{0, 1, 2\}$	Structure level of our team
$\mathcal{C} = (f, \zeta, \mathcal{G}, \sigma, r)$	Match configuration
r, ρ_j	Vision radius of a normal player; of opponent j
$\sigma \in \{0, \dots, 5\}$	Opponent playmaker index (0: none)
d_{sh}	Shadowing distance
R_{zone}, ω	Zone radius and ball-side pull of the zonal defence
μ_i^t, ν, ϖ	Belief over the opponent formation; assumed switch hazard; filter noise scale
χ	Individual shaping term (passes, tackles, shots)
m	Persistence length in detection delay
$\theta, \hat{\theta}$	Shared policy parameters; parameters returned by training
G_T	Final goal difference (ours minus opponent's)

with $c_{\min} \in (0, 1)$, $L > 0$. Control ratings scale tackle success, pass accuracy and shot success. The opponent always uses $\zeta = 2$. When a team switches formation, anchors change immediately and players must walk to them, so effectiveness drops temporarily.

2.3 Limited Vision and the Zonal Opponent

For a player at x with radius ρ , the **visibility indicator** of $z \in \mathcal{P}$ is $v(z; x, \rho) = \mathbb{1}[\|z - x\| \leq \rho]$. Normal players use $\rho = r$, and the playmaker (index σ) uses $\rho = \infty$. Opponents and the ball are observed only when $v = 1$; teammates are always visible (a simplification). A defender can attempt an interception only if the ball was visible to it when the pass was played.

Opponent j has home $\alpha_{g^t}(j)$ and reacts only to what it sees. With $u_j^t = \mathbb{1}[v(b^t; y_j^t, \rho_j) = 1]$, its desired position is

$$p_j^t = \begin{cases} b^t, & u_j^t = 1, \|b^t - \alpha_{g^t}(j)\| \leq R_{\text{zone}} \quad (\text{press}), \\ \alpha_{g^t}(j) + \omega \Pi_{R_{\text{zone}}}(b^t - \alpha_{g^t}(j)), & u_j^t = 1 \text{ otherwise} \quad (\text{drift}), \\ \alpha_{g^t}(j), & u_j^t = 0 \quad (\text{hold}), \end{cases} \quad (2)$$

where Π_R projects onto the disc of radius R and $\omega \in [0, 1]$. Excursions from home are bounded, so opponent positions stay informative about g^t , which makes the inference problem of Q2 well posed.

2.4 Dec-POMDP, Policies and Belief Filtering

A **Dec-POMDP** [2, 3] is a tuple $\langle \mathcal{N}, \mathcal{S}, \{\mathcal{A}_i\}, P, R, \{\Omega_i\}, O, \gamma, T \rangle$ with states, per-agent actions, transition kernel, shared team reward, observation spaces and kernel, and discount $\gamma \in (0, 1]$. Since the opponent is a fixed heuristic, we treat it as part of the environment.

The **history** of player i is $h_i^t = (o_i^0, a_i^0, \dots, o_i^t)$. A **shared policy** is $\pi_\theta(a_i^t | h_i^t, i)$; the recurrent version uses a hidden state $z_i^t = \text{GRU}_\theta(z_i^{t-1}, o_i^t, a_i^{t-1})$ [4]. Under **CTDE** [5], a critic $V_\phi(s^t)$ sees the full state in training, while at execution each player acts from its own history.

A **belief** [6] is a distribution over a hidden quantity. Let $\mu_i^t(g)$ be player i 's belief that the opponent formation is $g \in \mathcal{F}$. With switch hazard $\nu \in (0, 1)$, the **Bayesian filter** updates

$$\tilde{\mu}_i^t(g) = (1 - \nu) \mu_i^{t-1}(g) + \nu/|\mathcal{F}|, \quad (3)$$

$$\mu_i^t(g) \propto \tilde{\mu}_i^t(g) \prod_{j: v_{ij}^t=1} \mathcal{N}(y_j^t; \alpha_g(j), \varpi^2 I), \quad (4)$$

over the opponents visible to player i , with slots identified by player number and ϖ absorbing the pull of pressing and drift (bounded by R_{zone}). The hyperparameters (ν, ϖ) are chosen on validation matches from the *training* distribution, never the test distribution of Section 3.3.

2.5 Reward Shaping and Performance Measures

A shaping term $F(s, s')$ preserves the optimal policy if potential-based, $F = \gamma\Phi(s') - \Phi(s)$ [7]. Our shaping χ is **not**, so we bound and anneal it (Section 3.1). For configuration $\mathcal{C}$ and parameters θ , the **score** (draws count half) is

$$W(\theta; \mathcal{C}) = \Pr[G_T > 0] + \frac{1}{2} \Pr[G_T = 0], \quad (5)$$

over the environment, opponent and policy randomness. Effect sizes are differences of scores (win-rate points), each with a confidence interval.

3 Problem Formulation

3.1 Match Model

A **match configuration** is $\mathcal{C} = (f, \zeta, \mathcal{G}, \sigma, r)$: our formation, structure level, opponent formation process, playmaker index and vision radius.

State. $s^t = (x_{1:5}^t, y_{1:5}^t, b^t, \beta^t, g^t, t)$.

Actions. $a_i^t \in \mathcal{A}$: movement, hold-shape (only if $\zeta \geq 1$), pass-to-teammate, dribble, shoot, tackle, and *shadow-opponent- j* for each j , with invalid actions masked. Shadowing moves toward the point at distance d_{sh} from y_j^t on the line to our goal, and is masked unless $v_{ij}^t = 1$.

Observation.

$$o_i^t = \left(x_{1:5}^t, \beta^t, i, \{(y_j^t, v_{ij}^t)\}_{j \in \mathcal{M}}, (b^t, v_{ib}^t), \alpha_f(i) \mathbb{I}[\zeta \geq 1] \right), \quad (6)$$

with $v_{ij}^t = v(y_j^t; x_i^t, r)$, $v_{ib}^t = v(b^t; x_i^t, r)$, and hidden entries zeroed. The formation g^t , the switching mode and σ are *not* observed. Opponent j uses $\rho_j = \infty$ if $j = \sigma$ and $\rho_j = r$ otherwise.

Reward and objective.

$$R^t = \underbrace{\mathbb{I}[\text{goal for us}] - \mathbb{I}[\text{goal against}]}_{R_{\text{team}}^t} + \lambda_t \chi(s^t, \mathbf{a}^t) + \mathbb{I}[t = T] \eta \text{sign}(G_T), \quad J(\theta; \mathcal{C}) = \mathbb{E}_{\pi_{\theta, \mathcal{C}}} \left[\sum_{t=0}^T \gamma^t R^t \right], \quad (7)$$

with annealed weight $\lambda_t \rightarrow 0$, outcome bonus $\eta > 0$, and per-episode cap $\sum_t \lambda_t |\chi| \leq C$.

3.2 Q1: Can Formation Structure Compensate for Limited Vision?

Let $g^t \equiv g$ and $\sigma = 0$. PPO returns an approximate maximizer, so for each (ζ, f, r) we write

$$\hat{\theta}_{\zeta}(f, r) \approx \arg \max_{\theta} J(\theta; (f, \zeta, g, 0, r)), \quad W_{\zeta}(f, r) = W(\hat{\theta}_{\zeta}(f, r); (f, \zeta, g, 0, r)), \quad (8)$$

and conclusions concern trained policies, not exact optima. The **structure benefit** is $B_{\zeta}(f, r) = W_{\zeta}(f, r) - W_0(f, r)$ for $\zeta \in \{1, 2\}$, and the **best formation** is $f^*(r) = \arg \max_f W_2(f, r)$. We ask: is $B_{\zeta} > 0$ and decreasing in r ; does $f^*(r)$ change with r ; and does knowing the anchor ($\zeta = 1$) or being constrained to it ($\zeta = 2$) matter? As benchmark, let $M(f, f')$ be the Phase 0 heuristic win matrix. The **best-response** formation is $\arg \max_f M(f, g)$ and the **maximin** formation is $\arg \max_f \min_{f'} M(f, f')$; both are compared with $f^*(r)$.

3.3 Q2: Inferring and Adapting to a Formation Change

Switching process. $g^t = g_k$ for $t \in [\tau_k, \tau_{k+1})$, with switch times $\tau_1 < \tau_2 < \dots$ and targets $g_k \in \mathcal{F}$. Three modes define $\mathcal{G}$: *scheduled* (one switch at fixed τ_1 to a fixed target); *random* ($\tau_{k+1} - \tau_k \geq \delta$, g_k uniform on $\mathcal{F}$); *score-reactive* ($g_k = B(\Delta G^t)$ at check times, mapping a trailing opponent to an attacking formation and a leading one to a defensive one, with minimum dwell time δ).

Policy classes (all $\zeta = 2$, $\sigma = 0$): (1) *memoryless* $\pi_{\theta}(a \mid o_i^t, i)$; (2) *recurrent* $\pi_{\theta}(a \mid z_i^t, i)$, which must infer g^t implicitly; (3) *belief-augmented* $\pi_{\theta}(a \mid o_i^t, \mu_i^t, i)$ with the filter above, using no learning for inference; (4) *oracle* $\pi_{\theta^{\text{or}}}(a \mid o_i^t, g^t, i)$.

Training and testing. Train on $\mathcal{D}_2^{\text{train}}$ over $(\mathcal{G}, r)$, $\theta \in \arg \max_{\theta} \mathbb{E}_{(\mathcal{G}, r) \sim \mathcal{D}_2^{\text{train}}} [J(\theta; (f, 2, \mathcal{G}, 0, r))]$, and test on $\mathcal{D}_2^{\text{test}}$ whose switch times and (from, to) formation pairs are disjoint from training. The **regret** is $\text{Reg}(\theta) = W(\theta^{\text{or}}; \mathcal{C}) - W(\theta; \mathcal{C})$.

Detection delay. The inferred formation is $\hat{g}_i^t = \arg \max_g \mu_i^t(g)$ for the filter, and a probe $q_{\psi}(\hat{g}^t \mid z_i^t)$ on the hidden state for the recurrent policy. A switch counts as *detected* only when the inference is correct for m consecutive steps before the next switch:

$$\Delta_{\text{det}}(k) = \min \left\{ t - \tau_k : t \geq \tau_k, t + m - 1 < \tau_{k+1}, \hat{g}^{t'} = g_k \forall t' \in \{t, \dots, t + m - 1\} \right\}, \quad (9)$$

with undetected switches right-censored at $\tau_{k+1} - \tau_k$.

Adaptation lag. Goals are rare, so goal difference is averaged over N evaluation matches: with $G_t^{(n)}$ the goal difference in match n , $\bar{G}_{[a,b]} = \frac{1}{N(b-a)} \sum_n (G_b^{(n)} - G_a^{(n)})$. For window w and tolerance $\varepsilon \geq 0$,

$$\Delta_{\text{ad}}(k) = \min \left\{ t - \tau_k : t \geq \tau_k, \bar{G}_{[t, t+w]} \geq \bar{G}_{[\tau_k - w, \tau_k]} - \varepsilon \right\}, \quad (10)$$

with matches aligned at their own switch times. The same definition is applied to shot-difference rate and possession share as a denser secondary signal.

Objectives: estimate $\mathbb{E}[\Delta_{\text{det}}]$, $\mathbb{E}[\Delta_{\text{ad}}]$ and $\text{Reg}(\theta)$ per policy class, switching mode and radius; the key comparison is whether learned (recurrent) inference matches explicit (belief) inference, and how far both are from the oracle.

3.4 Q3: Adapting to an Opponent Playmaker

One opponent $\sigma \in \{1, \dots, 5\}$ has $\rho_{\sigma} = \infty$: he sees the ball, all players and all passing lanes, intercepts anywhere, and uses this view for his attacking decisions. The index $\sigma = 0$ means none, and his slot type (defender, midfielder, forward) is varied. Our team has no such player.

Policies. All share the same recurrent architecture and action set (including shadow actions), differing only in training and inputs: θ_{base} (trained with $\sigma = 0$ only, unaware); θ_{mix} (trained with $\sigma \sim \mathcal{D}_{\sigma}$ over $\{0, \dots, 5\}$, the adaptive team); θ_{or} (like θ_{mix} but given σ , the oracle). An *ablation* θ_{ns} is trained like θ_{mix} without shadow actions, to isolate their value.

Cost, recovery, regret, shadow value.

$$E(\sigma) = W(\theta_{\text{base}}; (f, 2, g, 0, r)) - W(\theta_{\text{base}}; (f, 2, g, \sigma, r)), \quad \kappa(\sigma) = \frac{W(\theta_{\text{mix}}; \sigma) - W(\theta_{\text{base}}; \sigma)}{E(\sigma)}, \quad (11)$$

$$\text{Reg}_{\sigma} = W(\theta_{\text{or}}; \sigma) - W(\theta_{\text{mix}}; \sigma), \quad A_{\text{sh}}(\sigma) = W(\theta_{\text{mix}}; \sigma) - W(\theta_{\text{ns}}; \sigma), \quad (12)$$

where $\kappa = 1$ is full recovery. Because κ divides by $E(\sigma)$, it is reported only when the confidence interval of $E(\sigma)$ excludes zero; otherwise the raw gain is reported.

Identification and response. A probe $q_{\psi}(\hat{\sigma}^t \mid z_i^t)$ on the hidden state of θ_{mix} has accuracy Acc_{σ} and **identification delay** $\Delta_{\text{id}} = \min\{t : \hat{\sigma}^{t'} = \sigma \forall t' \in \{t, \dots, T\}\}$. The **shadow rate** $\xi_{\sigma} = \frac{1}{T} \sum_t \mathbb{1}[\text{some player of ours shadows opponent } \sigma \text{ at } t]$ measures whether the team targets the playmaker; it is compared across θ_{mix} , θ_{or} and the case $\sigma = 0$ (where a high rate is wasted effort).

Objectives: estimate E , κ , Reg_{σ} , A_{sh} , Acc_{σ} , Δ_{id} and ξ_{σ} , and their dependence on the playmaker's position and on r .

4 Proposed Solutions

4.1 Environment and Heuristic Opponent (Phase 0)

Implement the 2D environment of Section 3.1 (continuous positions, ground-ball passing with interception geometry, dribbling, tackling, distance-based shooting, control rating, limited vision) and the heuristic opponent (zonal defence as in Section 2.3; pass to a free teammate ahead, shoot if unmarked in the shooting zone, otherwise dribble), with the playmaker variant and the three switching modes. **Calibrate** all constants by heuristic-vs-heuristic matches. **Phase 0 checks:** (i) a random policy loses clearly; (ii) vision matters (a small-radius team loses to the same team with a large radius) and the opponent's formation is readable from where it stands; (iii) a team with a full-vision playmaker beats the same team without one, so Q3 has a real effect to measure; (iv) the formation win matrix M with its best responses and maximin formation, the benchmark for Q1.

4.2 Learning Algorithm

Parameter-shared PPO [8] with generalized advantage estimation [9], a centralized critic and decentralized actors [10, 5], in feed-forward and GRU [4] actor variants; the belief-augmented actor is the feed-forward actor with the filter output appended. The reward has the two-tier form of Section 3.1 with annealed, capped shaping, and a short diagnostic run checks that shaping return does not grow while win rate stays flat. A curriculum starts with one formation and the largest radius, then all formations, then smaller radii. We use multiple training seeds and fixed evaluation seeds shared across policies for paired comparison.

4.3 Experiments, Expected Results and Hypotheses

Q1. Train a policy for each $\zeta \in \{0, 1, 2\}$, formation and radius against the fixed-formation zonal opponent. Report win rate versus radius per structure level, $B_\zeta(f, r)$ with confidence intervals, $f^*(r)$ against the best-response and maximin formations, and behaviour statistics (possession, passes, shots, spread). *Hypothesis:* structure helps more at small radii because a default position gives useful behaviour when a player sees little; knowing the anchor ($\zeta = 1$) may capture most of the benefit. Either outcome is informative.

Q2. Train and test the memoryless, recurrent, belief-augmented and oracle policies against the three switching modes at several radii, with held-out switch times and formation pairs, tuning (ν, ϖ) on training-distribution validation matches only. Report W and Reg per (mode, policy), Δ_{det} and Δ_{ad} per mode and radius (goal-based and shot/possession-based), and goal-difference curves aligned at τ_k . *Hypotheses:* recurrent beats memoryless; the belief filter is a strong baseline the recurrent policy may or may not match; the gap to the oracle grows as r shrinks.

Q3. Train θ_{base} , θ_{mix} , θ_{or} and θ_{ns} and evaluate them with the playmaker at a defender, midfielder and forward, at several radii. Report $E(\sigma)$, $\kappa(\sigma)$, Reg_σ , $A_{\text{sh}}(\sigma)$, Acc_σ , Δ_{id} , ξ_σ , and behaviour (does the adaptive team intercept more of his passes and give him less time on the ball). *Hypothesis:* the playmaker costs the unaware team measurable win rate, more so at small r ; the adaptive team recovers part of it but stays below the oracle; identification is easier when his long passes and distant interceptions reveal him.

4.4 Evaluation Protocol

- **Provisional settings (may change after Phase 0):** three formations (for example 2-2-1, 1-3-1, 3-1-1) and three vision radii plus full vision.
- **Seeds and matches:** 3 training seeds on the full Q1 grid ($3 \times 3 \times 4 \times 3 = 108$ runs) and 5 seeds on headline conditions only; 300 or more evaluation matches per condition and seed, since 100 matches give a standard error near 0.05 that hides effects of a few points. Evaluation is cheap compared with training.
- Mean and standard deviation over seeds, paired evaluation seeds across policies, and confidence intervals on every effect size.
- Always run the random-action baseline first, and track shaping return against win rate as the reward-hacking indicator.
- A negative or partial result is valid if quantified (for example, “structure gives no benefit at small r ”).

4.5 Priorities, Scope, Success Criteria and Challenges

Priority: Phase 0 and Q1 are the core, then Q2, then Q3 (extra training setting and shadow actions). If time runs short: drop the no-shadow ablation, then reduce playmaker positions, then formations, and cut headline runs to 3 seeds. **Out of scope:** a learned coach choosing formations, substitutions and fatigue, learning opponents (self-play), and other defensive styles such as man-marking (which would make formation inference much harder).

Success criteria. Q1: a quantified answer, with confidence intervals, on whether the structure benefit and best formation change with the radius. Q2: measured detection delay, adaptation lag and regret per policy class and mode, including a clear recurrent-versus-filter comparison. Q3: the measured playmaker cost, the fraction recovered by the adaptive team, and its gap to the oracle. A well-quantified negative result counts as success.

Challenges and mitigations. *Simulator effort:* minimal mechanics and an early Phase 0. *Opponent quality:* the Phase 0 checks. *Reward hacking and instability:* annealed and capped shaping, curriculum, monitoring. *Compute:* the Q1 grid alone is over a hundred runs, so 3 seeds on the grid, 5 on headline conditions, and reuse of earlier policies. *Weak or noisy effects:* 300 or more matches, effect sizes with confidence intervals, and shot and possession rates as a denser signal than sparse goals. *Simplifications:* teammates always visible, slots identified by number, fixed opponent, radius-based vision; conclusions are limited to this setting.

5 Contribution

Table 2: Proposed task assignment.

Member	Responsibility	Tasks
Arpan Gupta	Environment and opponent (Sec. 4.1)	2D environment (passing, interception, dribbling, tackling, shooting, control rating, limited vision); zonal opponent, playmaker variant, shadow action, three switching modes; Phase 0 calibration and checks, including the formation win matrix with best-response and maximin formations.
Deep Buha	Learning and Q1 (Secs. 4.2, 4.3)	Parameter-shared PPO with centralized critic (feed-forward and GRU); reward design with annealed, capped shaping and the reward-hacking monitor; structure levels and the Q1 experiments.
Ramji Purwar	Inference and playmaker, Q2 and Q3 (Secs. 4.3, 4.3, 4.4)	Bayesian belief filter (tuned on training data only), belief-augmented and oracle variants, probes; Q2 experiments (switching modes, held-out schedules, delay, lag, regret); Q3 experiments (cost, recovery, identification, shadow behaviour, ablation); evaluation pipeline with confidence intervals, tables and figures.

References

- [1] Hiroaki Kitano, Minoru Asada, Yasuo Kuniyoshi, Itsuki Noda, and Eiichi Osawa. RoboCup: The robot world cup initiative. In *Proceedings of the First International Conference on Autonomous Agents*, 1997.
- [2] Daniel S. Bernstein, Robert Givan, Neil Immerman, and Shlomo Zilberstein. The complexity of decentralized control of Markov decision processes. *Mathematics of Operations Research*, 27(4), 2002.
- [3] Frans A. Oliehoek and Christopher Amato. *A Concise Introduction to Decentralized POMDPs*. Springer, 2016.
- [4] Kyunghyun Cho et al. Learning phrase representations using RNN encoder-decoder for statistical machine translation. In *Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2014.
- [5] Ryan Lowe et al. Multi-agent actor-critic for mixed cooperative-competitive environments. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.
- [6] Leslie Pack Kaelbling, Michael L. Littman, and Anthony R. Cassandra. Planning and acting in partially observable stochastic domains. *Artificial Intelligence*, 101(1–2), 1998.
- [7] Andrew Y. Ng, Daishi Harada, and Stuart Russell. Policy invariance under reward transformations: Theory and application to reward shaping. In *Proceedings of the International Conference on Machine Learning (ICML)*, 1999.
- [8] John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347*, 2017.
- [9] John Schulman, Philipp Moritz, Sergey Levine, Michael Jordan, and Pieter Abbeel. High-dimensional continuous control using generalized advantage estimation. In *International Conference on Learning Representations (ICLR)*, 2016.
- [10] Chao Yu et al. The surprising effectiveness of PPO in cooperative multi-agent games. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.
- [11] Karol Kurach et al. Google research football: A novel reinforcement learning environment. In *Proceedings of the AAAI Conference on Artificial Intelligence*, 2020. arXiv:1907.11180.
- [12] Yan Song, He Jiang, Haifeng Zhang, Zheng Tian, Weinan Zhang, and Jun Wang. Boosting studies of multi-agent reinforcement learning on google research football environment: the past, present, and future. *arXiv preprint arXiv:2309.12951*, 2023.